ZCAS University

CCS2412 – COMPUTER ORGANIZATION AND ARCHITECTURE

END OF SEMESTER EXAMINATION

Friday, 6th December 2024

09:00 HRS-12:00HRS

TIME ALLOWED: **WRITING – THREE HOURS**

READING TIME – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Numbering of your answers in your booklet should be the same as that in the question paper. Answers that are not numbered will NOT be marked.
5. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

Efficient interaction between the CPU, Memory, and Storage is a cornerstone of computer organization and architecture. Understanding how these components function and relate is critical for optimizing performance in computer systems. With that in mind, you are required to answer the following questions:

i) Define the concept of memory hierarchy in computer systems. **(3 Marks)**

ii) Explain how the memory hierarchy impacts the overall performance of a computer system. **(3 Marks)**

iii) Describe the relationship between memory hierarchy levels (Registers, Cache, RAM, and Secondary Storage) and their impact on system performance. **(12 Marks)**

iv) Explain why a multilevel cache system (L1, L2, L3) improves performance compared to a single-level cache. **(4 Marks)**

v) Explain why a balance between memory cost and performance is essential in designing a computer system. **(6 Marks)**

vi) Explain the principle of storage in a Solid-State Drive (SSD) and why it is preferred over Hard Disk Drive (HDD) for high-performance systems. **(6 Marks)**

vii) State one benefit and one drawback associated with hybrid storage systems that combine Solid State Drives (SSDs) and Hard Disk Drives (HDDs) in a single computer system. **(6 Marks)**

(Total: 40 Marks)

SECTION B: Attempt any THREE questions in this section

Question 2

Storage Management:

i) Tasila has a laptop with a total storage capacity of 1TB. She plans to store 100 high-resolution videos on her laptop. On average, each video is 4GB in size. How much storage space (in MB) will remain unused after storing the videos? [Show the working] **(4 Marks)**

ii) A USB drive has a capacity of 32GB. It is currently 75% full of data. How much data (in KB) is still available on the USB drive for new files? [Show the working] **(4 Marks)**

iii) A cloud storage provider offers a 10TB subscription plan. A user uploads 2,500 high-definition videos, each averaging 2GB in size. How much storage space (in GB) remains after uploading the videos? [Show the working] **(4 Marks)**

iv) A database server stores 10 million records. If each record is 150KB in size, what is the total storage space (in gigabytes) is required for these records? [Show the working] **(4 Marks)**

v) A movie file is 2GB in size. How many bits are in this file? [Show the working] **(4 Marks)**

(Total: 20 Marks)

Question 3

MIPS stands for Microprocessor without Interlocked Pipeline Stages. It is an example of an instruction set architecture (ISA). There are 3 types of Instruction Formats in the MIPS ISA Assembly Language:

- R-Instruction Format
- I-Instruction Format
- J-Instruction Format

In the MIPS assembly language, the I-Instruction Format is typically used for instructions that perform operations with an immediate value (a constant or offset) and one or more registers. An example of an instruction in I-Instruction Format is given below:

lw $t0, 20($s1)

i) From the above instruction, you are required to:
   a) Identify and state the Operand(s);
   b) Identify and state the Opcode(s);
   c) Identify and state the Destination Register(s);
   d) Identify and state the Source Register(s);
   e) Explain the role of the $ symbol **(10 Marks)**

ii) With the MIPS ISA Assembly Language in mind:
   a) Explain the steps which will be followed when the above instruction is executed. **(3 Marks)**
   b) Can the format below be accepted in MIPS ISA? Explain why it would or why it would not be accepted. lw $t1, 20($s0) **(3 Marks)**

iii) In MIPS ISA, the load (lw) command and the store (sw) command are both used to access memory. Using any appropriate instruction for each command, you are required to explain the difference between the two commands in terms of usage. **(4 Marks)**

(Total: 20 Marks)

Question 4

The CPU, or Central Processing Unit, is the primary hardware component of a computer responsible for executing instructions. It is the "brain" of the computer and performs arithmetic and logic operations, data processing, and control tasks. It has the following major components:

- Control Unit (CU)
- Arithmetic and Logic Unit (ALU)
- Registers
- Buses

i) What is the role of the Control Unit? **(2 Marks)**

ii) What is the role of the Arithmetic and Logic Unit? **(2 Marks)**

iii) What is the role for each of the following Registers:
   a) Accumulator (ACC)
   b) Program Counter (PC)
   c) Memory Address Register (MAR)
   d) Memory Data Register (MDR)
   e) Current Instruction Register (CIR) **(10 Marks)**

209715 2 KB

iv) What is the role for each of the following Buses:

a) Data Bus

b) Address Bus

c) Control Bus **(6 Marks)**

(Total: 20 Marks)

Question 5

Data representation. ... Computers use binary - the digits 0 and 1 - to store data. A binary digit, or bit, is the smallest unit of data in computing. It is represented by a 0 or a 1. Binary numbers are made up of binary digits (bits). The circuits in a computer's processor are made up of billions of transistors. Digits / Characters, music files, video files and any other data type that is processed by a computer, is stored using their binary equivalent. The actual processing is done by the Central Processing Unit through the use of primary memory devices.

i) ASCII and Unicode are among the commonly used standard codes for characters in computer systems. State and explain any difference between the two coding systems. **(3 Marks)**

ii) Using a computer system that utilises an ASCII Coding System, a student typed the following characters and saved them "Data." The binary representation (bytes) associated to the typed characters on her computer is as given below:

D	a	t	a	.
01000100	01100001	01110100	01100001	00101110

Use any method to find the binary representation (bytes) for the characters below, including a full stop / period. Show your working clearly or the logic you used to arrive at the answers. **(7 Marks)**

A	r	e	.

iii) Use any method to determine the **byte representation** of the integer 83 in the computer system. [Show the Working] **(2 Marks)**

iv) Use any method to determine the **equivalent integer** for the 01101001 byte which is currently stored in one of the computer memories. [Show the Working] **(2 Marks)**

v) The sentence below was typed by a student in notepad and saved on the desktop of her laptop that uses an ASCII coding system.

Attending classes is a must do for students.

The sentence above is made of 44 characters including spaces and a full stop. Analyze the sentence and answer the following questions:

a) What is the number of bits that make up the sentence? [Show the Working]
(2 Marks)

b) Indicate the size of the notepad file in bytes. Explain how you arrive at your answer.
(2 Marks)

c) What would be the size of the file in bytes under the Unicode environment? Explain how you arrive at your answer. **(2 Marks)**

(Total: 20 Marks)

END OF EXAMINATION